

Towards automated screening and coding for infectious disease systematic reviews with large language models

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Abstract

Systematic reviews underpin parameterization of infectious disease models but are limited by labor-intensive screening of large literature corpora with low inclusion yield. Although large language models (LLMs) show promise for automation, their performance in literature screening remains suboptimal without domain-specific design. Here, we present a dimension-aware LLM framework for literature screening, evaluating articles across three epidemiological characteristics: transmission potential, transmission speed, and disease severity. Using a benchmark of 22 COVID-19 and mpox systematic review profiles (12,758 records; 668 included studies), the model achieves consistently high recall across parameters: 0.984 (95% CI: 0.959–0.994) for reproduction number, 0.935 (95% CI: 0.894–0.961) for serial interval, and 0.957 (95% CI: 0.919–0.977) for fatality; and reduces manual screening workload by 60% (95% CI: 0.59–0.62), 83% (95% CI: 0.82–0.84), and 76% (95% CI: 0.75–0.77), respectively. These findings demonstrate that dimension-aware LLMs can substantially improve screening efficiency while preserving high sensitivity, supporting their role as assistive tools in human-in-the-loop systematic review workflows.

1 Introduction

Accurate epidemiological characteristics are essential for evidence-based public health decision-making during infectious disease outbreaks.^{1–4} Parameters such as the basic reproduction number (R_0), serial interval, and case fatality rate serve as quantitative indicators of transmission potential, transmission speed, and disease severity, respectively. These parameters critically inform the development of public health strategies, the design of intervention measures, and the allocation of healthcare resources. However, obtaining reliable estimates of these parameters requires rigorous evidence synthesis.

Although parameter estimates from single high-impact studies or preliminary reports are commonly used, they often introduce substantial uncertainty and propagate systematic bias throughout model simulations, ultimately undermining the reliability and interpretability of projections. Systematic reviews and meta-analyses offer a superior alternative as the most authoritative source of such parameters. Through standardized integration of evidence across studies, they effectively reduce biases related to publication, geography, and population heterogeneity.^{5–7} Only by adopting parameters derived from such comprehensive evidence synthesis can transmission models move beyond theoretical constructs to deliver credible, policy-relevant predictions.

Nevertheless, despite generating high-quality evidence, the systematic review entails substantial resource requirements and considerable operational complexity. An empirical investigation of PROSPERO-registered projects

has demonstrated that the average duration from protocol initiation to full publication of a systematic review extends to 67.3 weeks, about 1.29 years.⁸ Moreover, the literature screening process is particularly labor-intensive: initial database searches frequently retrieve tens of thousands of records, ranging from 27 to over 90,000, of which only about 2.94% typically meet the prespecified eligibility criteria.⁸ Consequently, researchers must devote considerable effort to screening vast numbers of citations to identify a small set of eligible studies.

In response to this labor-intensive bottleneck, artificial intelligence has emerged as a promising solution. Recent advances in artificial intelligence (AI) have enabled AI-assisted tools to alleviate the burden of manual screening, with active learning algorithms demonstrating particular promise in prioritizing relevant citations while maintaining high sensitivity.^{9–11} Nevertheless, existing unmodified large language models remain incapable of independently executing the critical task of identifying and selecting relevant literature for systematic reviews. This limitation is starkly illustrated by the performance of three fully automated GPT-4-based approaches: Consensus GPT, Scholar GPT, and the native GPT web-browsing mode correctly identified merely 32.7% (32/98), 19.4% (19/98), and 6.1% (6/98) of eligible articles, respectively.¹² These poor results highlight the need for domain-specific adaptation. While specialized models such as MedCPT in biomedical information retrieval¹³ and TrialMind in clinical trial screening¹⁴ have shown substantially better performance, the field of AI-assisted systematic review still lacks a dedicated, high-performance retrieval framework.

Here, we present MetaAgent, a dimension-aware large language model framework developed for high-recall literature screening in systematic reviews of infectious disease modeling parameters. The framework evaluates each candidate article across five clinically and methodologically relevant dimensions: disease relevance, population relevance, location relevance, original empirical evidence, and parameter relevance; and employs a stage-aware screening strategy that adapts the input modality according to the availability of title and abstract information. By integrating these design elements, MetaAgent aims to preserve high sensitivity while substantially reducing the manual screening burden. This study demonstrates that dimension-aware LLMs can serve as effective assistive tools within human-in-the-loop systematic review workflows, providing a practical pathway to accelerate high-quality evidence synthesis for infectious disease modeling and public health decision-making.

2 Method

We developed a framework named MetaAgent for automated literature screening of systematic reviews in infectious disease modeling (Figure 1). The proposed MetaAgent consists of three core components: benchmark construction, LLM screening, and evaluation, which jointly support the literature screening workflow for relevant systematic reviews.

2.1 Benchmark Dataset

This study adopts three core epidemiological parameters to evaluate the proposed screening framework: basic reproduction number (R_0), serial interval, and fatality, characterizing the transmission potential, transmission speed, and disease severity of an epidemic. We constructed a screening benchmark from published systematic reviews on COVID-19/SARS-CoV-2 and mpox. The benchmark was frozen on May 8, 2026. The benchmark comprises 22 screening profiles, each corresponding to one systematic review task, with a total of 12,758 candidate records. The source reviews used to define the benchmark are listed in Table S6. For each task, we designed a Boolean search query based on the target parameter definitions, adhering as closely as possible to the original search strategies reported in the source reviews. PubMed served as the primary data source for retrieving records, which collectively formed the raw pool of candidate articles for screening.

Upon constructing the raw pool, we defined the final inclusion list of each systematic review as the corresponding ground truth (GT) for that task. In this framework, the raw pool delineates the search space of candidate records, whereas the GT identifies the subset of articles that should ultimately be included. Across the benchmark, the aggregated GT consists of 668 articles. It is important to emphasize that the GT in this study does not constitute a

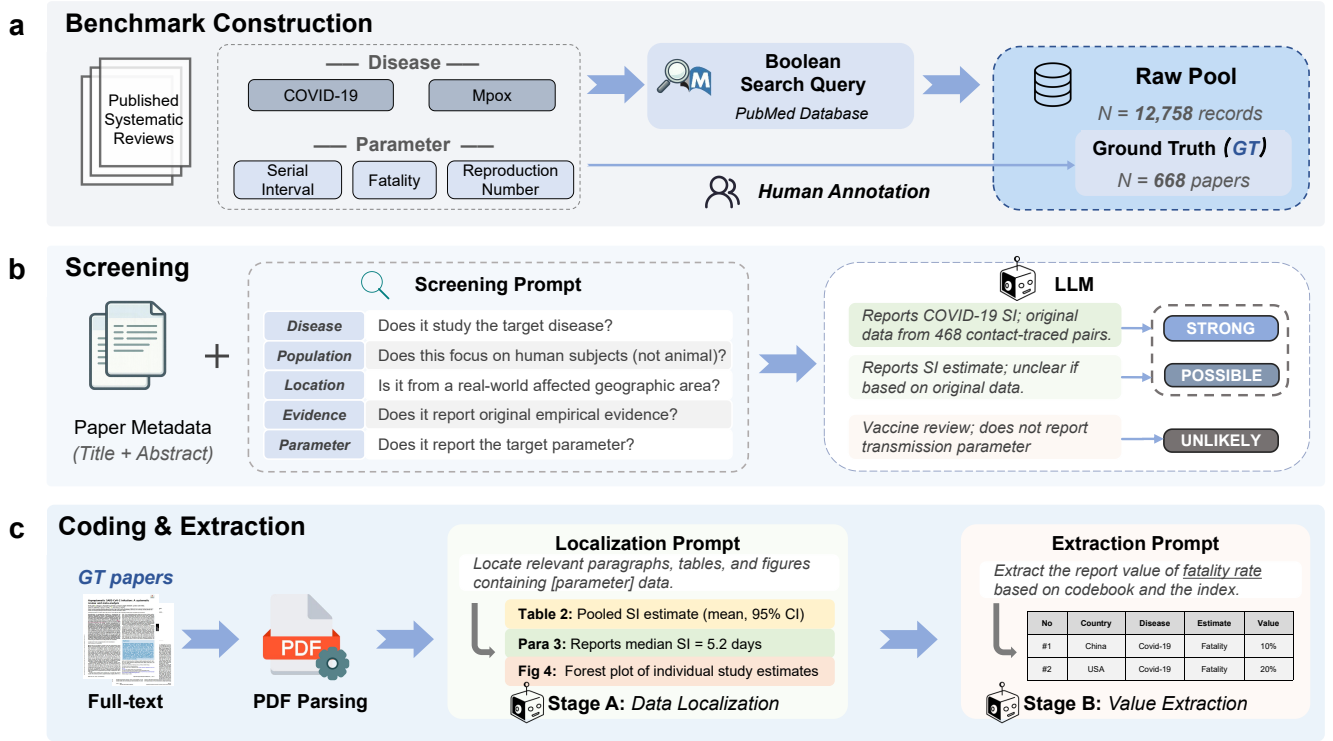


Figure 1: Overview of the MetaAgent pipeline. a) Benchmark construction from published systematic reviews. b) LLM screening of title and abstract. c) Full-text PDF parsing and codebook-guided parameter extraction.

rigorous gold standard in the strictest sense. The original search strategies often relied on multiple bibliographic databases and could not be fully reconstructed from the published manuscripts alone, meaning that some GT articles may not appear in our constructed raw pool. The original reviews may themselves contain retrieval omissions or include eligibility constraints that are not fully encoded in the current tasks.

2.2 LLM Screening

Task definition. We formulate the task as high-recall first-pass screening, rather than a definitive inclusion/exclusion decision. The objective of the system is to maximally retain GT-positive articles while substantially reducing the number of records requiring manual review, thereby focusing human effort on studies with a higher likelihood of relevance. Final inclusion judgments remain the responsibility of domain experts operating at the full-text level.

Screening strategy. We adopt a stage-aware screening strategy that tailors the input modality to the level of metadata available for each candidate record. This design acknowledges that the completeness of bibliographic metadata varies across records retrieved from PubMed, and that treating all records uniformly risks either discarding potentially relevant studies prematurely or introducing noise into the downstream decision process.

- **Title+Abstract screening.** For records that contain a usable abstract, the LLM receives both the title and the abstract as input. This richer representation allows the model to assess the five scoring dimensions with greater contextual fidelity, yielding more reliable dimension-level scores and a more informed relevance categorization (Strong, Possible, or Unlikely).
- **Title-only screening.** For records that lack an available or informative abstract, such as certain preprint entries, letters, or older publications with missing metadata, the LLM is constrained to a title-only input. Recognizing the inherent uncertainty introduced by this reduced information signal, the title-only pathway applies the same five-dimension scoring rubric but is explicitly designed to err on the side of retention.

Records classified as Possible or Strong via title-only screening proceed to the next stage, whereas those deemed Unlikely are excluded.

By conditioning the screening depth on metadata availability, the stage-aware strategy balances efficiency with the overarching high-recall objective of the first-pass screening task: to preserve as many GT-positive articles as possible while substantially reducing the volume of records requiring downstream human review.

LLM screening. The input to the large language model consists of two components: (1) paper metadata for each candidate article in the raw pool, including title, abstract, and keywords; and (2) a screening prompt that operationalizes five independent scoring dimensions:

- Disease relevance: the study must explicitly target the infectious disease of interest.
- Population relevance: the study should concern human subjects or populations. Purely animal or in vitro studies are considered less relevant.
- Location relevance: the study should ideally be situated in a real-world geographic or field context.
- Original empirical evidence: the study should report original empirical data or primary analysis. Reviews, editorials, commentaries, and purely theoretical papers are deemed less relevant.
- Parameter relevance: the study should report or estimate the target epidemiological parameter.

Each dimension is scored independently, and the resulting scores are subsequently combined through a threshold-based mapping to assign each record to one of three relevance categories: Strong, Possible, or Unlikely.

Score-to-label mapping. Based on the scores assigned across the five dimensions, each article is mapped into one of three relevance categories: Strong, Possible, or Unlikely. The Strong label designates records that receive high scores on critical dimensions, such as disease, parameter, and evidence type. The Possible label is reserved for records that are likely relevant but for which the metadata alone does not support a high-confidence determination. The Unlikely label is assigned to records that fail to meet the specified thresholds.

- **Strong:** disease relevance ≥ 4 , parameter relevance ≥ 4 , original empirical evidence ≥ 3 , population relevance ≥ 3 , and location relevance ≥ 3 .
- **Possible:** disease relevance ≥ 3 , parameter relevance ≥ 3 , original empirical evidence ≥ 2 , population relevance ≥ 3 , and location relevance ≥ 3 .
- **Unlikely:** any record that does not meet the criteria for either Strong or Possible.

2.3 Coding and Extraction

Task definition. We evaluated full-text coding as a controlled evidence-reconstruction task (Figure 1c). For this component, the input was fixed to the papers included by each source review. The objective was not to make a new eligibility decision, but to recover the parameter evidence required to reproduce the review-level epidemiological estimate. This design isolates the coding step from title-and-abstract screening, so that discrepancies in the reconstructed estimates primarily reflect full-text availability, document parsing, evidence localization, value extraction, or alignment with the source review’s synthesis rule.

Full-text processing. For each source-review included article, cached full texts were used when available; otherwise, open-access full text was retrieved with PubMed Central prioritized where possible. PDFs were parsed into a structured text representation that retained section order, paragraphs, figure captions, and table-like content where recoverable. The parsed representation was cached before LLM coding, ensuring that repeated experiments used the same document input.

Evidence localization and extraction. The coding workflow followed the two-stage structure shown in the method overview. Stage A localized candidate evidence windows, including paragraphs, tables, and figure captions that mentioned the target parameter, sample size, denominator, population, study period, geography, uncertainty interval, or estimation method. Stage B applied a parameter-specific codebook to these localized windows and produced schema-constrained records. Each record captured the parameter type, point estimate, uncertainty interval, sample size or numerator-denominator pair when available, study setting, subgroup definition, and source location in the full text.

Separate codebooks were used for the three parameter domains. Reproduction-number coding distinguished R_0 , effective reproduction number, and other transmissibility estimates. Transmission-interval coding distinguished serial interval, generation interval or generation time, and incubation period. Fatality coding distinguished CFR, IFR, hospitalized fatality, ICU fatality, and denominator-specific or clade-specific estimates. These distinctions were necessary because the same article can report multiple quantities that are not interchangeable for meta-analysis.

Estimate reconstruction. The structured coding outputs were aggregated at the source-review level and compared with the estimates reported by the original reviews. For continuous parameters, reported uncertainty intervals were converted to standard errors when possible and combined using inverse-variance random-effects synthesis. For fatality proportions, reconstructed estimates were interpreted only when the denominator, population stratum, and source-review aggregation rule could be aligned. Rows based on medians, narrative ranges, aggregate counts, or clade-specific summaries were therefore evaluated against the corresponding source-review estimand rather than forced into a single pooling rule.

2.4 Evaluation

2.4.1 Screening

The screening performance of the model was evaluated by comparing its outputs against the manually annotated GT. For evaluation purposes, both Strong and Possible relevance judgments were consolidated into a single relevant category, while Unlikely was treated as not relevant. Based on this binary classification, the following four outcomes were defined: true positives (TP), articles classified as relevant that were included in the ground truth; false positives (FP), articles classified as relevant but not included in the ground truth; false negatives (FN), articles classified as not relevant but included in the ground truth; and true negatives (TN), articles classified as not relevant and not included in the ground truth.

From these four categories, we computed recall, precision, workload reduction (WR), and number needed to screen (NNS). Recall measures the proportion of GT-relevant articles within the raw pool that were correctly identified by the system. Precision describes the proportion of articles advanced to manual review that are genuinely relevant, thereby quantifying the false-positive burden imposed by a high-recall strategy. WR denotes the proportion of articles classified as not relevant by the system and consequently exempted from manual review. NNS represents the average number of articles that must be manually reviewed per eligible study identified; a lower NNS indicates greater screening efficiency. Given that the proposed framework is designed as a high-recall screening tool for the title-and-abstract phase, recall and WR were regarded as the two most decision-informative metrics in the present experiments.

2.4.2 Coding

Coding performance was evaluated as agreement between reconstructed estimates and source-review reference estimates. We reported absolute differences in point estimates and whether the reconstructed estimate was closer to the source-review estimate than the LEADS estimate. Summary agreement was evaluated overall, by disease, and by parameter domain. This validation should be interpreted as component-level coding validity under a fixed source-review inclusion set, not as end-to-end performance from search retrieval to final meta-analysis.

3 Results

3.1 LLM-Assisted Screening Performance

MetaAgent was evaluated on a benchmark comprising 22 published COVID-19 and mpox systematic review profiles, covering 12,758 candidate records and 668 ground-truth included studies. The benchmark spans three key epidemiological parameters: reproduction number, serial interval, and fatality; which correspond to transmission potential, transmission speed, and disease severity, respectively. Candidate records were retrieved from PubMed, and the final inclusion lists reported in the original systematic reviews were used as task-specific ground truth. MetaAgent performs stage-aware screening using available title and abstract information, and assigns each article to one of three relevance categories: Strong, Possible, or Unlikely. For evaluation, Strong and Possible records were treated as relevant and advanced to manual review.

Across all tasks, MetaAgent achieved an overall recall of 0.960 (95% CI: 0.942–0.972) while reducing the manual screening workload by 73.6% (95% CI: 0.728–0.744). Parameter-specific analyses showed consistently high recall, reaching 0.984 (95% CI: 0.959–0.994) for reproduction number, 0.935 (95% CI: 0.894–0.961) for serial interval, and 0.957 (95% CI: 0.919–0.977) for fatality. These results demonstrate that MetaAgent can substantially reduce the number of articles requiring human screening while preserving high sensitivity in systematic review workflows. Model-level comparisons and per-baseline screening metrics are reported in Figure S1 and Table S1.

We evaluated MetaAgent’s performance on literature screening tasks based on the benchmark dataset. The system achieved a high overall recall of 0.960, with the highest recall of 0.984 (95% CI: 95.9%, 99.4%) observed in the reproduction number group. The fatality group ranked second in terms of recall performance, reaching 0.957 (0.919, 0.977). By contrast, the serial interval group yielded the lowest recall at 0.935 (0.894, 0.961). This indicates that approximately 96% of truly relevant articles in the raw pool could be retained for subsequent manual review.

The number of FNs was small across all groups. Taking the serial interval group as an example, the 14 FNs primarily originated from records where the target parameter only appeared in the full-text tables or appendices but not in the abstracts. This pattern reflects the inherent information limitation of relying solely on abstracts for the first-stage screening, rather than systematic misjudgment by the model. Thus, it represents an expected and acceptable source of error in the methodological design. A focused full-text tier-2 ablation for the Ali et al. serial-interval review is provided in Table S2, with remaining false-positive categories summarized in Table S3.

The system achieved an overall WR rate of 73.6%, meaning that approximately three-quarters of candidate articles could be automatically excluded without proceeding to manual review. The serial interval group had the highest WR rate of 83% and an NNS of only 2.55, indicating that reviewers needed to screen an average of approximately 3 articles to confirm 1 positive article. In contrast, due to the extremely low density of the candidate pool in the fatality group, even though the WR rate reached 76%, the NNS remained as high as 7.47. This suggests that the screening efficiency is ultimately constrained by parameter specificity rather than the discriminative ability of the model itself.

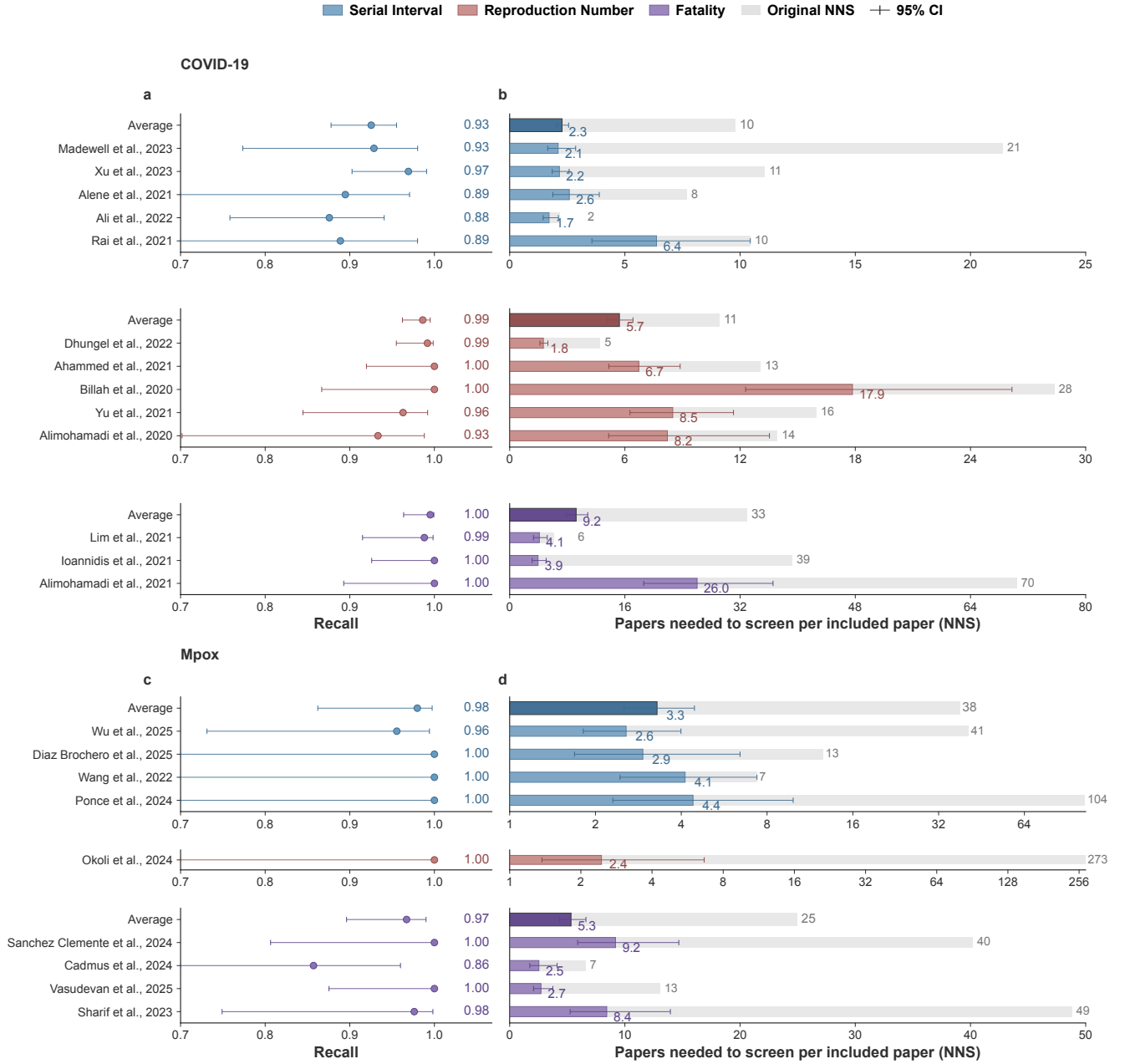


Figure 2: LLM-assisted screening recall and NNS (papers needed to screen per included paper) across COVID-19 and mpox benchmarks. a) and c) per-review recall grouped by parameter domain (colour-coded); filled circles are per-review recall; horizontal bars are Wilson 95% CIs. b) and d) corresponding NNS; coloured bars = LLM-assisted, grey bars = original unassisted; whiskers are 95% CIs.

3.2 LLM-Coded Estimate Reconstruction

We next evaluated the full-text coding component under a controlled setting in which the eligible articles were fixed to the inclusion lists of the source reviews. The analysis therefore tested whether the Stage A localization and Stage B extraction workflow could recover the evidence needed to reconstruct review-level parameter estimates, independent of upstream screening errors.

Across 22 source reviews, the reconstructed estimates closely matched the source-review references (Figure 3). Agreement was strongest when the review estimand and pooling rule were explicit, such as serial-interval and reproduction-number analyses. For COVID-19, reconstructed serial-interval estimates differed from the source-

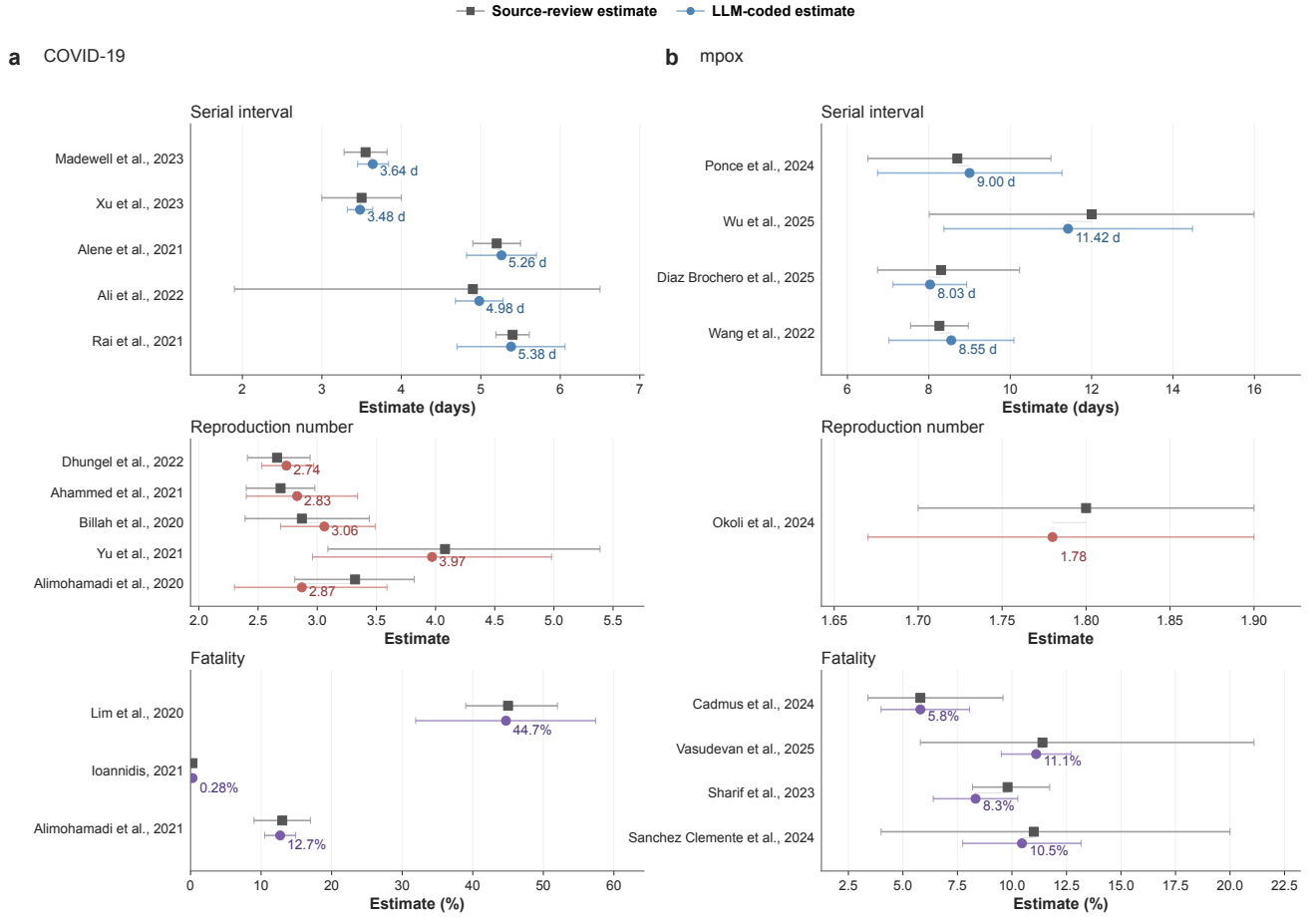


Figure 3: Agreement between LLM-coded estimates and source-review estimates. Panels a and b show source-review and LLM-coded estimates for serial interval, reproduction number, and fatality in COVID-19 and mpox reviews. Points denote point estimates and horizontal bars denote 95% confidence intervals. Grey squares indicate estimates reported in the source reviews; coloured circles indicate estimates reconstructed from LLM-coded full-text evidence. Axes are scaled separately by disease and parameter.

review estimates by 0.02–0.09 days in four of five reviews, and reproduction-number estimates preserved the same review-level ordering as the original analyses. Fatality estimates were also recovered when the denominator was aligned to the source review, including invasive-mechanical-ventilation fatality, hospitalized fatality, and mpox clade- or population-specific CFR.

The comparison with LEADS supported the same conclusion. Overall, our coded estimates had lower mean absolute error than LEADS (0.255 versus 2.677) and were closer to the source-review estimate in 21 of 22 source reviews (Table S4). This advantage was observed in both diseases, with closer estimates in 13 of 13 COVID-19 reviews and 8 of 9 mpox reviews. The only mpox exception was a transmission-interval row where both methods remained close to the source-review estimate. These results indicate that codebook-guided full-text coding can reproduce review-level epidemiological estimates when the target parameter definition and synthesis rule are made explicit. Full per-review comparisons are shown in Figure S2 and Table S5.

4 Discussion

Systematic reviews could provide the evidentiary foundation for characterizing infectious disease epidemics, yet their production remains constrained by the labor-intensive process of literature screening. In this study, we developed MetaAgent, a dimension-aware large language model framework designed for high-recall first-pass screening of studies relevant to infectious disease modeling parameters. Across 22 COVID-19 and mpox

systematic review profiles comprising 12,758 candidate records and 668 ground-truth included studies, MetaAgent achieved an overall recall of 0.960 while reducing manual screening workload by 73.6%. Performance was consistent across three epidemiological parameter domains, with recall reaching 0.984 for reproduction number, 0.935 for serial interval, and 0.957 for fatality. These findings suggest that dimension-aware LLM screening can substantially reduce the number of records requiring human review while preserving the sensitivity required in systematic review workflows.

A key contribution of MetaAgent is its explicit alignment with the screening logic used by domain experts. Rather than relying on a single binary relevance judgment, MetaAgent decomposes the screening task into five interpretable dimensions: disease relevance, population relevance, location relevance, original empirical evidence, and parameter relevance. This design reflects the fact that eligibility in infectious disease parameter reviews is rarely determined by one criterion alone. For example, an article may focus on the correct disease but report no empirical estimate of the target parameter, or it may estimate a relevant parameter using non-human or purely theoretical data. By scoring these dimensions separately and mapping them to Strong, Possible, and Unlikely categories, MetaAgent provides a transparent intermediate representation of relevance, supporting human oversight and reducing the risk of premature exclusion.

The framework was intentionally optimized for recall rather than precision. In systematic review screening, false negatives are generally more consequential than false positives, because missed eligible studies may bias downstream evidence synthesis and parameter estimation. MetaAgent therefore treats both Strong and Possible articles as relevant for manual review. This conservative strategy explains the moderate overall precision but is appropriate for a first-pass screening tool whose purpose is not to replace expert judgment, but to prioritize records and reduce the screening burden. This is particularly critical in resource-constrained emergency settings, such as guiding evidence-based public health responses during sudden outbreaks. The observed workload reduction indicates that a large proportion of clearly irrelevant records can be excluded automatically, allowing reviewers to focus on a smaller and more enriched set of candidate studies.

Performance varied across parameter domains, reflecting differences in how epidemiological concepts are reported in the literature. Fatality-related reviews achieved the highest recall, suggesting that severity outcomes may be more explicitly described in titles and abstracts. Serial interval reviews showed lower recall, and inspection of false negatives indicated that some eligible studies reported the target parameter only in full-text tables, appendices, or secondary analyses rather than in the abstract. This limitation is inherent to title-and-abstract screening and is not unique to LLM-based methods. Reproduction number tasks showed high recall but lower workload reduction than serial interval tasks, likely because broad mentions of transmissibility and epidemic growth make it more difficult to distinguish articles that estimate reproduction numbers from those that merely discuss them. These findings highlight that screening difficulty is partly determined by the semantic specificity of the target parameter.

MetaAgent also addresses an important limitation of generic LLM applications in systematic review automation. Previous fully automated GPT-based approaches have shown limited ability to identify eligible articles in systematic reviews, underscoring the need for domain-specific task design. In contrast, MetaAgent does not require domain-specific model retraining, but improves screening performance through structured prompting, dimension-level reasoning, and stage-aware input handling. This suggests that carefully designed LLM workflows may provide practical gains even without fine-tuning, particularly when the task can be decomposed into expert-defined criteria. Such a design may be especially valuable in rapidly evolving outbreak contexts, where timely evidence synthesis is needed but labeled training data may be limited.

Despite these strengths, several limitations should be considered. First, the ground truth was derived from the inclusion lists of published systematic reviews rather than from a newly adjudicated gold standard. Published reviews may contain retrieval omissions, apply eligibility criteria that are incompletely reported, or include studies identified through databases and supplementary searches not reproduced in our benchmark. As a result, recall and precision estimates should be interpreted as performance against review-derived ground truth, not as definitive measures against an exhaustive corpus of all eligible studies. Second, our benchmark was con-

structured primarily from PubMed, whereas the original systematic reviews often searched multiple bibliographic databases. This structural asymmetry may limit the generalizability of the reported performance to complete real-world systematic review workflows involving Embase, Web of Science, Scopus, preprint servers, and manual citation tracking.

MetaAgent provides valuable assistance to medical experts and systematic reviewers by streamlining the literature mining process and offering superior temporal efficiency compared to purely manual efforts. However, the feasibility of AI integration remains constrained by several unresolved concerns, including algorithmic opacity, compromised reproducibility across temporally distinct queries, and limited capacity for context-dependent interpretation.²⁸ Thus, while AI offers considerable potential as an adjunctive instrument for accelerating review workflows, a critical examination of its methodological boundaries and the enduring necessity of human oversight warrants further investigation.^{29,30}

Despite these strengths, several limitations should be considered. First, the ground truth was derived from the inclusion lists of published systematic reviews rather than from a newly adjudicated gold standard. Published reviews may contain retrieval omissions, apply eligibility criteria that are incompletely reported, or include studies identified through databases and supplementary searches not reproduced in our benchmark. As a result, recall and precision estimates should be interpreted as performance against review-derived ground truth, not as definitive measures against an exhaustive corpus of all eligible studies. Second, our benchmark was constructed primarily from PubMed, whereas the original systematic reviews often searched multiple bibliographic databases. This structural asymmetry may limit the generalizability of the reported performance to complete real-world systematic review workflows involving Embase, Web of Science, Scopus, preprint servers, and manual citation tracking. Third, full-text tier-2 screening and coding were evaluated as controlled downstream components rather than as a fully prospective end-to-end review. Articles labeled Strong or Possible still require expert review, and final inclusion decisions remain the responsibility of human reviewers. Fourth, the method depends on the information available in titles, abstracts, accessible full texts, and parsed tables. Studies that report relevant parameters only in supplementary materials, images, or poorly parsed tables may still be missed or incompletely coded. Incorporating broader full-text retrieval and multimodal table understanding may reduce this source of error, but would introduce additional challenges related to access, document parsing, and computational cost.

More broadly, integrating LLMs into evidence synthesis raises methodological and operational concerns. LLM outputs may vary across model versions, prompts, and time, affecting reproducibility. The internal reasoning process of proprietary models may not be fully transparent, and model behavior may shift as systems are updated. These issues are particularly important for systematic reviews, where reproducibility and auditability are central methodological principles. The dimension-aware scoring design of MetaAgent improves interpretability at the decision level, but does not eliminate the need for careful documentation, version control, prompt archiving, and human verification.

Future work should evaluate MetaAgent prospectively in live systematic review projects and across a wider range of pathogens, study designs, and epidemiological parameters. Multi-database benchmarks would better reflect real-world search workflows, while prospective validation could assess how the system performs when screening newly published literature not available during model pretraining. Further development should also explore integration with full-text screening, citation chasing, uncertainty calibration, and active-learning prioritization. Combining LLM-based semantic assessment with conventional information retrieval methods may further improve both sensitivity and efficiency.

In summary, MetaAgent demonstrates that dimension-aware LLM screening can serve as an effective assistive tool for systematic reviews of infectious disease modeling parameters. By preserving high recall while substantially reducing manual workload, the framework offers a practical approach to accelerating evidence synthesis without removing human experts from the decision-making process. These results support the use of structured, human-in-the-loop LLM systems as complementary infrastructure for systematic reviews, particularly in public health settings where timely and reliable parameter evidence is essential.

References

1. IHME COVID-19 Forecasting Team. Modeling COVID-19 scenarios for the United States. *Nat. Med.*, 27: 94–105, 2021.
2. C. J. E. Metcalf and J. Lessler. Opportunities and challenges in modeling emerging infectious diseases. *Science*, 357:149–152, 2017.
3. S. Cobey. Modeling infectious disease dynamics. *Science*, 368:713–714, 2020.
4. G. L. Vasconcelos et al. Power law behaviour in the saturation regime of fatality curves of the COVID-19 pandemic. *Sci. Rep.*, 11:4619, 2021.
5. A. P. Siddaway, A. M. Wood, and L. V. Hedges. How to do a systematic review: A best practice guide for conducting narrative reviews, meta-analyses, and meta-syntheses. *Annu. Rev. Psychol.*, 70:747–770, 2019.
6. L. J. Jensen, J. Saric, and P. Bork. Literature mining for the biologist: from information retrieval to biological discovery. *Nat. Rev. Genet.*, 7:119–129, 2006.
7. Z. Wang et al. A foundation model for human-AI collaboration in medical literature mining. *Nat. Commun.*, 16:8361, 2025.
8. R. Borah, A. W. Brown, P. L. Capers, and K. A. Kaiser. Analysis of the time and workers needed to conduct systematic reviews of medical interventions using data from the PROSPERO registry. *BMJ Open*, 7:e012545, 2017.
9. R. Ofori-Boateng, M. Aceves-Martins, N. Wiratunga, and C. F. Moreno-Garcia. Towards the automation of systematic reviews using natural language processing, machine learning, and deep learning: a comprehensive review. *Artif. Intell. Rev.*, 57:131, 2024.
10. X. Hou et al. Large language models for software engineering: a systematic literature review. *ACM Trans. Softw. Eng. Methodol.*, 33:1–79, 2024.
11. D. Scherbakov, N. Hubig, V. Jansari, A. Bakumenko, and L. A. Lenert. The emergence of large language models as tools in literature reviews: a large language model-assisted systematic review. *J. Am. Med. Inform. Assoc.*, 32:1071–1086, 2025.
12. H. Chen et al. Can large language models fully automate or partially assist paper selection in systematic reviews? *Br. J. Ophthalmol.*, 109:962–966, 2025.
13. Q. Jin et al. MedCPT: Contrastive pre-trained transformers with large-scale PubMed search logs for zero-shot biomedical information retrieval. *Bioinformatics*, 39:btad651, 2023.
14. Z. Wang et al. Accelerating clinical evidence synthesis with large language models. *NPJ Digit. Med.*, 8:509, 2025.
15. Z. J. Madewell et al. Rapid review and meta-analysis of serial intervals for SARS-CoV-2 Delta and Omicron variants. *BMC Infect. Dis.*, 23:429, 2023.
16. X. Xu et al. Assessing changes in incubation period, serial interval, and generation time of SARS-CoV-2 variants of concern: a systematic review and meta-analysis. *BMC Med.*, 21:374, 2023.
17. M. Alene et al. Serial interval and incubation period of COVID-19: a systematic review and meta-analysis. *BMC Infect. Dis.*, 21:257, 2021.
18. S. T. Ali et al. Serial intervals and case isolation delays for Coronavirus disease 2019: a systematic review and meta-analysis. *Clin. Infect. Dis.*, 74:685–694, 2022.

19. B. Rai, A. Shukla, and L. K. Dwivedi. Estimates of serial interval for COVID-19: a systematic review and meta-analysis. *Clin. Epidemiol. Glob. Health*, 9:157–161, 2021.
20. M. A. Billah, M. M. Miah, and M. N. Khan. Reproductive number of coronavirus: a systematic review and meta-analysis based on global level evidence. *PLoS One*, 15:e0242128, 2020.
21. C.-J. Yu et al. Assessment of basic reproductive number for COVID-19 at global level: a meta-analysis. *Medicine (Baltimore)*, 100:e25837, 2021.
22. Y. Alimohamadi, M. Taghdir, and M. Sepandi. Estimate of the basic reproduction number for COVID-19: a systematic review and meta-analysis. *J. Prev. Med. Public Health*, 53:151–157, 2020.
23. B. Dhungel et al. Reliability of early estimates of the basic reproduction number of COVID-19: a systematic review and meta-analysis. *Int. J. Environ. Res. Public Health*, 19:11613, 2022.
24. T. Ahammed et al. Estimation of novel coronavirus (COVID-19) reproduction number and case fatality rate: a systematic review and meta-analysis. *Health Sci. Rep.*, 4:e274, 2021.
25. Z. J. Lim et al. Case fatality rates for patients with COVID-19 requiring invasive mechanical ventilation: a meta-analysis. *Am. J. Respir. Crit. Care Med.*, 203:54–66, 2021.
26. J. P. A. Ioannidis. Infection fatality rate of COVID-19 inferred from seroprevalence data. *Bull. World Health Organ.*, 99:19–33F, 2021.
27. Y. Alimohamadi, H. H. Tola, A. Abbasi-Ghahramanloo, M. Janani, and M. Sepandi. Case fatality rate of COVID-19: a systematic review and meta-analysis. *J. Prev. Med. Hyg.*, 62:E311–E320, 2021.
28. M. Moens, G. Nagels, N. Wake, and L. Goudman. Artificial intelligence as team member versus manual screening to conduct systematic reviews in medical sciences. *iScience*, 28:113559, 2025.
29. C. Galli, A. V. Gavrilova, and E. Calciolari. Large language models in systematic review screening: opportunities, challenges, and methodological considerations. *Information (Basel)*, 16:378, 2025.
30. R. Sanghera et al. High-performance automated abstract screening with large language model ensembles. *J. Am. Med. Inform. Assoc.*, 32:893–904, 2025.
31. S. Wang et al. Serial intervals and incubation periods of the monkeypox virus clades. *J. Travel Med.*, 29:taac105, 2022.
32. L. Ponce et al. Incubation period and serial interval of mpox in 2022 global outbreak compared with historical estimates. *Emerg. Infect. Dis.*, 30:1173–1181, 2024.
33. S. Wu et al. Global transmission characteristics of mpox outbreaks: a systematic review and meta-analysis. *EClinicalMedicine*, 89:103526, 2025.
34. C. Diaz Brochero et al. Decoding mpox: a systematic review and meta-analysis of the transmission and severity parameters of the 2022–2023 global outbreak. *BMJ Glob. Health*, 10:e016906, 2025.
35. G. N. Okoli et al. A global systematic evidence review with meta-analysis of the epidemiological characteristics of the 2022 mpox outbreaks. *Infection*, 52:901–921, 2024.
36. S. Cadmus et al. Dynamics of mpox infection in Nigeria: a systematic review and meta-analysis. *Sci. Rep.*, 14:7368, 2024.
37. S. S. Vasudevan et al. Global and regional mortality rate statistics of mpox (formerly monkeypox): a comprehensive systematic review and meta-analysis. *Pathog. Glob. Health*, 119:293–303, 2025.

38. N. Sharif et al. Molecular epidemiology, transmission and clinical features of 2022-mpox outbreak: a systematic review. *Health Sci. Rep.*, 6:e1603, 2023.
39. N. Sanchez Clemente et al. Paediatric, maternal, and congenital mpox: a systematic review and meta-analysis. *Lancet Glob. Health*, 12:e572–e588, 2024.

Appendix

Appendix A. Metric Definitions

For each model evaluated on the screening benchmark, the following metrics are reported. TP = true positives (included papers correctly retrieved); FP = false positives (excluded papers incorrectly retrieved); FN = false negatives (included papers missed); TN = true negatives (excluded papers correctly excluded).

$$\text{Recall} = \frac{TP}{TP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

$$\text{NNS} = \frac{TP + FP}{TP}, \quad \text{WR} = 1 - \frac{TP + FP}{TP + FP + FN + TN}.$$

Appendix B. Screening Model Comparison

Supplementary Figure S1 visualizes recall and papers needed to screen across candidate models and baselines. Table S1 reports the corresponding aggregate counts and metrics on the unified benchmark.

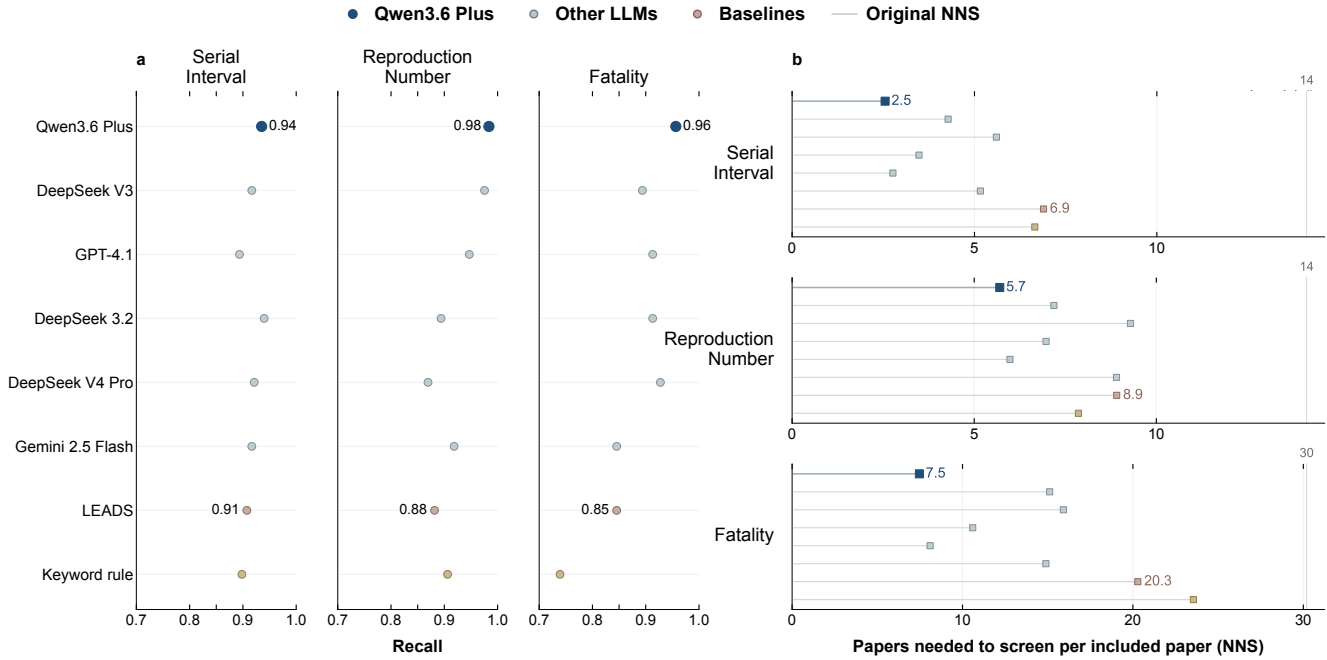


Figure S1: Model-assisted screening performance across epidemiological parameters. a) Recall across three parameter domains, shown as dot plots to emphasize small differences among high-recall models. b) Papers needed to screen per included paper (NNS), where lower values indicate fewer papers screened per included paper; numbers above each panel indicate the original unassisted NNS.

Table S1: Overall screening performance by model or baseline on the unified benchmark.

Model or method	Raw Pool	GT	TP	FP	FN	TN	Recall	Precision	F1	NNS	WR
Qwen3.6 Plus			641	2727	27	9363	0.960	0.190	0.318	5.25	0.736
DeepSeek V3			622	4739	46	7351	0.931	0.116	0.206	8.62	0.580
GPT-4.1			614	5631	54	6459	0.919	0.098	0.178	10.17	0.511
DeepSeek 3.2	12,758	668	611	3625	57	8465	0.915	0.144	0.249	6.93	0.668
DeepSeek V4 Pro			604	2775	64	9315	0.904	0.179	0.298	5.59	0.735
Gemini 2.5 Flash			598	5034	70	7056	0.895	0.106	0.190	9.42	0.559
LEADS			587	6239	81	5851	0.879	0.086	0.157	11.63	0.465
Keyword rule			569	6070	99	6020	0.852	0.086	0.156	11.67	0.480

Appendix C. Full-Text Tier-2 Ablation

Tables S2 and S3 summarize the full-text rescreening ablation and the remaining false-positive categories for the COVID-19 serial-interval review by Ali et al. 2022.

Table S2: Stage-2 full-text rescreening ablation for the COVID-19 serial-interval review by Ali et al. 2022. S, P, and U denote papers predicted at Stage 1 as strong, possible, and excluded, respectively; full texts were obtained from PMC OA where available.

Stage	Strategy	TP	FP	FN	TN	Recall	Precision	F1	NNS
Stage I (Title+Abstract)	-	45	40	6	20	0.882	0.529	0.662	1.89
Stage II (Full-text)	S-only	45	39	6	21	0.882	0.536	0.667	1.87
	P-only	45	25	6	35	0.882	0.643	0.744	1.56
	U-only	48	40	3	20	0.941	0.545	0.691	1.83
	S+P	45	24	6	36	0.882	0.652	0.750	1.53
	S+P+U	48	24	3	36	0.941	0.667	0.780	1.50

Table S3: Supplementary audit of remaining false positives after the S+P+U full-text tier-2 policy for the COVID-19 serial-interval review by Ali et al. 2022. Counts are against the current PMID-level source-review ground truth.

Remaining false-positive category	Count
Target-parameter positive but absent from GT	7
Outside source-review indexing window	5
Secondary synthesis/review	3
Methodological or modelling study	2
Effective/time-varying SI definition	1
Secondary model synthesis	1
Borderline extractability	1
Duplicate/version PMID mismatch	1
Correspondence or reanalysis	1
Early parameter summary	1
Borderline or absent from GT	1

Appendix D. Coding Results Comparison

Supplementary Figure S2 and Tables S4–S5 provide the full per-source-review coding comparison between Ours and LEADS.

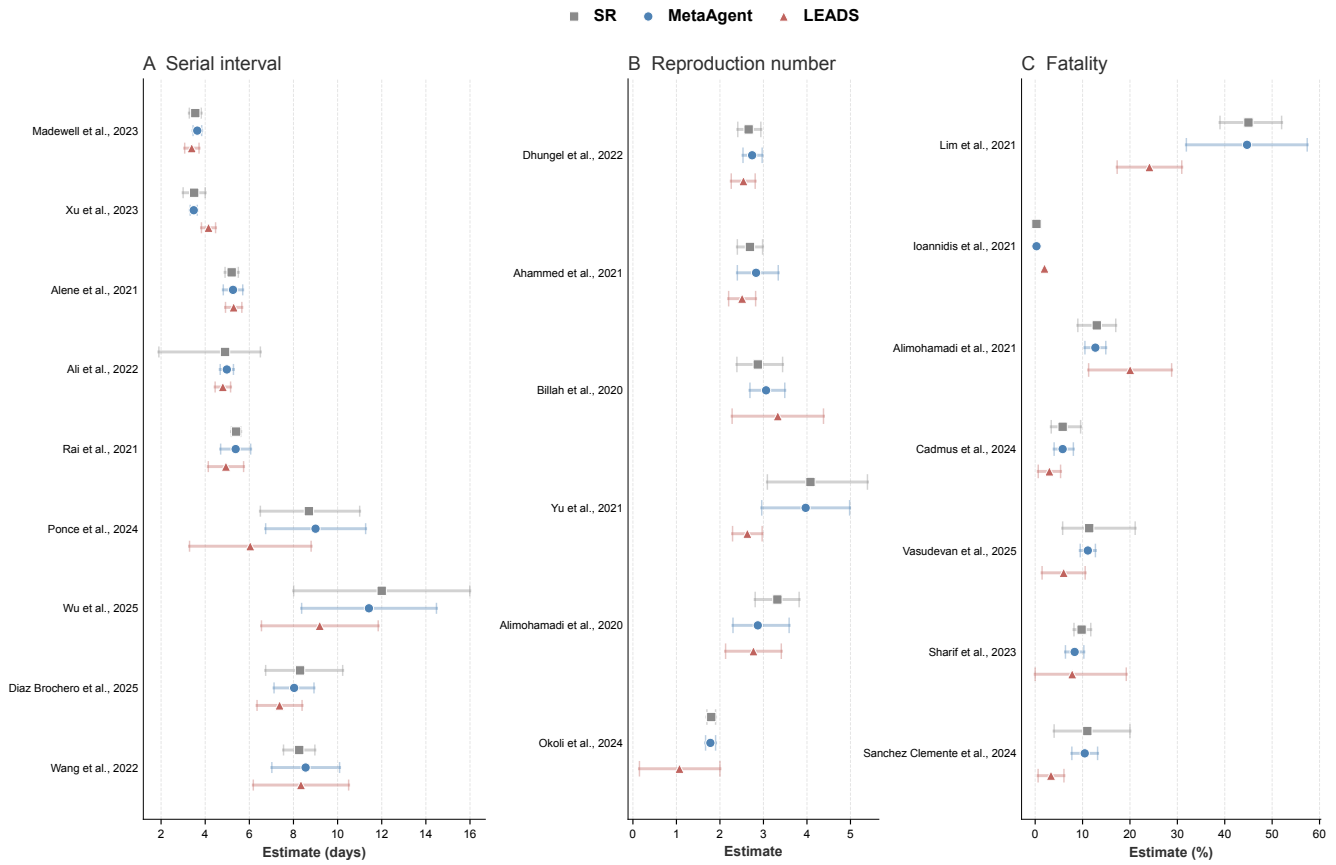


Figure S2: Supplementary coding comparison of source-review (SR), Ours, and LEADS estimates across serial interval, reproduction number, and fatality.

Table S4: Coding estimate agreement against source-review references.

Scope	Source reviews	Ours MAE	LEADS MAE	Ours better	LEADS better	LEADS n
Overall	22	0.255	2.677	21	1	625
Continuous estimates	15	0.180	0.762	14	1	426
Transmission interval	9	0.190	0.882	8	1	217
Reproduction number	6	0.165	0.582	6	0	209
Fatality	7	0.416	6.781	7	0	199
COVID-19	13	0.142	2.604	13	0	525
Mpox	9	0.418	2.783	8	1	100

Table S5: Per-source-review coding comparison of Ours and LEADS against source-review reference estimates.

Disease	Parameter	Source review	Estimate (95% CI)			Δ		Ours better?
			SR	Ours	LEADS	Ours	LEADS	
COVID-19	Serial Interval	Madewell et al., 2023	3.55 d (3.28–3.82)	3.64 d (3.45–3.84)	3.39 d (3.07–3.72)	0.09	0.16	✓
		Xu et al., 2023	3.50 d (3.00–4.00)	3.48 d (3.32–3.64)	4.15 d (3.83–4.47)	0.02	0.65	✓
		Alene et al., 2021	5.20 d (4.90–5.50)	5.26 d (4.82–5.70)	5.29 d (4.92–5.66)	0.06	0.09	✓
		Ali et al., 2022	4.90 d (1.90–6.50)	4.98 d (4.68–5.28)	4.80 d (4.45–5.15)	0.08	0.10	✓
		Rai et al., 2021	5.40 d (5.19–5.61)	5.38 d (4.70–6.06)	4.94 d (4.14–5.74)	0.02	0.46	✓
	Reproduction Number	Dhungel et al., 2022	2.66 (2.41–2.94)	2.74 (2.53–2.97)	2.54 (2.26–2.81)	0.08	0.12	✓
		Ahammed et al., 2021	2.69 (2.40–2.98)	2.83 (2.40–3.34)	2.51 (2.20–2.82)	0.14	0.18	✓
		Billah et al., 2020	2.87 (2.39–3.44)	3.06 (2.69–3.49)	3.33 (2.28–4.38)	0.19	0.46	✓
		Yu et al., 2021	4.08 (3.09–5.39)	3.97 (2.96–4.98)	2.63 (2.29–2.97)	0.11	1.45	✓
		Alimohamadi et al., 2020	3.32 (2.81–3.82)	2.87 (2.30–3.59)	2.77 (2.13–3.41)	0.45	0.55	✓
	Fatality	Lim et al., 2021	45.00% (39.00–52.00)	44.70% (31.90–57.40)	24.11% (17.30–30.93)	0.30	20.89	✓
		Ioannidis et al., 2021	0.27%	0.28%	1.97%	0.01	1.70	✓
		Alimohamadi et al., 2021	13.00% (9.00–17.00)	12.70% (10.50–14.90)	20.04% (11.28–28.79)	0.30	7.04	✓
Mpox	Serial Interval	Wang et al., 2022	8.26 d (7.55–8.97)	8.55 d (7.02–10.09)	8.34 d (6.18–10.50)	0.29	0.08	×
		Ponce et al., 2024	8.70 d (6.50–11.00)	9.00 d (6.74–11.27)	6.04 d (3.29–8.80)	0.30	2.66	✓
		Wu et al., 2025	12.00 d (8.01–15.99)	11.42 d (8.37–14.48)	9.19 d (6.55–11.84)	0.58	2.81	✓
		Diaz Brochero et al., 2025	8.30 d (6.74–10.23)	8.03 d (7.12–8.93)	7.37 d (6.35–8.39)	0.27	0.93	✓
	Reproduction Number	Okoli et al., 2024	1.80 (1.70–1.90)	1.78 (1.67–1.90)	1.07 (0.15–2.00)	0.02	0.73	✓
	Fatality	Cadmus et al., 2024	5.80% (3.39–9.59)	5.80% (4.00–8.06)	3.00% (0.65–5.36)	0.00	2.80	✓
		Vasudevan et al., 2025	11.40% (5.80–21.10)	11.11% (9.51–12.71)	6.01% (1.46–10.55)	0.29	5.39	✓
		Sharif et al., 2023	9.80% (8.19–11.73)	8.33% (6.39–10.27)	7.81% (0.00–19.22)	1.47	1.99	✓
		Sanchez Clemente et al., 2024	11.00% (4.00–20.00)	10.46% (7.74–13.17)	3.34% (0.62–6.07)	0.54	7.66	✓

Appendix E. Benchmark Source-Review Inventory

Table S6 lists the source reviews used to construct the frozen screening and coding benchmark.

Table S6: Supplementary source-review inventory for the frozen screening and coding benchmark. Disease and parameter cells are grouped by source-review block.

Disease	Parameter	Source review	Raw pool	GT
COVID-19	Serial interval	Madewell et al., 2023	599	28
		Xu et al., 2023	841	76
		Alene et al., 2021	146	19
		Ali et al., 2022	111	51
		Rai et al., 2021	94	9
		<i>Subtotal</i>	1,791	183
	Reproduction number	Dhungel et al., 2022	574	122
		Ahammed et al., 2021	575	44
		Billah et al., 2020	709	25
		Yu et al., 2021	575	36
		Alimohamadi et al., 2020	209	15
		<i>Subtotal</i>	2,642	242
	Fatality	Lim et al., 2021	345	56
		Ioannidis et al., 2021	1,882	48
		Alimohamadi et al., 2021	2,253	32
		<i>Subtotal</i>	4,480	136
	Total		8,913	561
Mpox	Serial interval	Wang et al., 2022	59	8
		Ponce et al., 2024	522	5
		Wu et al., 2025	612	15
		Diaz Brochero et al., 2025	63	5
		<i>Subtotal</i>	1,256	33
	Reproduction number	Okoli et al., 2024	819	3
		Fatality	Cadmus et al., 2024	92
	Vasudevan et al., 2025		352	27
	Sharif et al., 2023		683	14
	Sanchez Clemente et al., 2024		643	16
	<i>Subtotal</i>		1,770	71
	Total		3,845	107
Overall		12,758	668	

Acknowledgements

To be completed.

Data Availability

Coming soon.

Code Availability

Coming soon.